4. MODES OF DES

AIM

To implement different modes of the Data Encryption Standard (DES) algorithm using Python.

Algorithm

1. Start the program.
2. Import the required DES library.
3. Enter the plaintext.
4. Enter an 8-character secret key.
5. Select the DES mode (ECB, CBC, CFB, OFB, or CTR).
6. Encrypt the plaintext.
7. Display the encrypted ciphertext.
8. Decrypt the ciphertext.
9. Display the original plaintext.
10. Stop.

(A) DES - ECB MODE

Python Code

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
import base64
plaintext = input("Enter Plain Text: ")
key = input("Enter 8-character Key: ").encode()
cipher = DES.new(key, DES.MODE_ECB)
ciphertext = cipher.encrypt(pad(plaintext.encode(), DES.block_size))
print("\nEncrypted Text :", base64.b64encode(ciphertext).decode())
plain = unpad(cipher.decrypt(ciphertext), DES.block_size)
print("Decrypted Text :", plain.decode())
```

Output

```
•••  Enter Plain Text: bhavya
     Enter 8-character Key: security

     Encrypted Text : 5zQFCF7KnWg=
     Decrypted Text : bhavya
```

(B) DES - CBC MODE

Python Code

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
from Crypto.Random import get_random_bytes
import base64
plaintext = input("Enter Plain Text: ")
key = input("Enter 8-character Key: ").encode()
iv = get_random_bytes(8)
cipher = DES.new(key, DES.MODE_CBC, iv)
ciphertext = cipher.encrypt(pad(plaintext.encode(), DES.block_size))
print("\nEncrypted Text :", base64.b64encode(ciphertext).decode())
cipher = DES.new(key, DES.MODE_CBC, iv)
plain = unpad(cipher.decrypt(ciphertext), DES.block_size)
print("Decrypted Text :", plain.decode())
```

Output

```
•••  Enter Plain Text: hi how are u
     Enter 8-character Key: security

     Encrypted Text : u3vgHg51ifvPpG6Pyy8W0w==
     Decrypted Text : hi how are u
```

(C) DES - CFB MODE

Python Code

```
from Crypto.Cipher import DES
from Crypto.Random import get_random_bytes
import base64
plaintext = input("Enter Plain Text: ")
key = input("Enter 8-character Key: ").encode()
iv = get_random_bytes(8)
cipher = DES.new(key, DES.MODE_CFB, iv)
ciphertext = cipher.encrypt(plaintext.encode())
print("\nEncrypted Text :", base64.b64encode(ciphertext).decode())
cipher = DES.new(key, DES.MODE_CFB, iv)
plain = cipher.decrypt(ciphertext)
print("Decrypted Text :", plain.decode())
```

Output

```
Enter Plain Text: hi i am fine
Enter 8-character Key: security

Encrypted Text : Q1A+cpuMyfFFMwT1
Decrypted Text : hi i am fine
```

(D) DES - OFB MODE

Python Code

```
from Crypto.Cipher import DES
from Crypto.Random import get_random_bytes
import base64
plaintext = input("Enter Plain Text: ")
key = input("Enter 8-character Key: ").encode()
iv = get_random_bytes(8)
cipher = DES.new(key, DES.MODE_OFB, iv)
```

```
ciphertext = cipher.encrypt(plaintext.encode())
print("\nEncrypted Text :", base64.b64encode(ciphertext).decode())
cipher = DES.new(key, DES.MODE_OFB, iv)
plain = cipher.decrypt(ciphertext)
print("Decrypted Text :", plain.decode())
```

Output

```
Enter Plain Text: hows ur mother
Enter 8-character Key: security

Encrypted Text : PVTJ5Vc3Vs7LDtKPHO4=
Decrypted Text : hows ur mother
```

(E) DES - CTR MODE

Python Code

```
from Crypto.Cipher import DES
from Crypto.Util import Counter
import base64
plaintext = input("Enter Plain Text: ")
key = input("Enter 8-character Key: ").encode()
ctr = Counter.new(64)
cipher = DES.new(key, DES.MODE_CTR, counter=ctr)
ciphertext = cipher.encrypt(plaintext.encode())
print("\nEncrypted Text :", base64.b64encode(ciphertext).decode())
ctr = Counter.new(64)
cipher = DES.new(key, DES.MODE_CTR, counter=ctr)
plain = cipher.decrypt(ciphertext)
print("Decrypted Text :", plain.decode())
```

Output

```
Enter Plain Text: my mother is fine
Enter 8-character Key: security

Encrypted Text : 6/l7Fn2gkFWOOyVa+HmV5nI=
Decrypted Text : my mother is fine
```

The result:

The above codes are executed and the output is obtained successfully.